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Quantum Rényi relative entropy and trace inequality

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Introduction

Entropy and relative entropy play an important role in information theory. Among different entropies, the Rényi entropy can be said one of the most important. It generalizes the concept of entropy and is found useful in many applications. For instances, Rényi relative entropy will reduces to relative entropy when α approaches to 1, min-entropy when α approaches to 0, and max-entropy when α approach to ∞ In this project, we will review an Rényi entropy called α-sandwiched Rényi relative entropy and in the end, we will introduce a new trace inequality based on the Rényi relative entropy.

1 Notations

- ρ, σ: Quantum states, which are described as positive semi-definite, self-adjoint, and has unit trace operator.
- $D(\rho,\sigma) := \operatorname{Tr}\rho(\log\rho - \log\sigma)$: Quantum relative entropy.

2 α-sandwiched Rényi relative entropy[1,2,3]

α-sandwiched Rényi relative entropy, or sometimes called sandwiched Rényi relative entropy, is defined as

$$\tilde{D}_\alpha(\rho||\sigma) = \frac{1}{\alpha-1}\log\operatorname{Tr}[(\sigma^{\frac{1-\alpha}{2\alpha}}\rho\sigma^{\frac{1-\alpha}{2\alpha}})^\alpha]$$

In many papers, the quantity is introduced with the help of sandwiched quasi-relative entropy[1,3] or some authors call it parametrized version of fidelity,[2] which is defined as

$$\tilde{Q}_\alpha(\rho||\sigma) = \operatorname{Tr}[(\sigma^{\frac{1-\alpha}{2\alpha}}\rho\sigma^{\frac{1-\alpha}{2\alpha}})^\alpha]$$

Then, we restated the sandwiched Rényi relative entropy as

$$\tilde{D}_\alpha(\rho||\sigma) = \frac{1}{\alpha-1}\log\tilde{Q}_\alpha(\rho||\sigma)$$

In this representation, we know that by analyzing the sandwiched quasi-relative entropy, we can get some properties of sandwiched Rényi relative entropy.

The following are some of the properties of sandwiched Rényi relative entropy listed in the paper[1,3]

- $\tilde{Q}_\alpha(\rho||\sigma)$ is invariant under all unitaries.
- $\tilde{Q}_\alpha$ is invariant under tensoring another quantum system.
- $\tilde{Q}_\alpha(\rho_1\otimes\rho_2||\sigma_1\otimes\sigma_2) = \tilde{Q}_\alpha(\rho_1||\sigma_1)\tilde{Q}_\alpha(\rho_2||\sigma_2)$
- $\tilde{Q}_\alpha(\rho||\sigma)$ is differentiable in α
- $\tilde{Q}_\alpha$ is jointly convex for $\alpha \in (1,2]$
- $\tilde{D}_\alpha(\rho||\sigma) \geq 0$, for $\alpha \in (1,2]$ and equality holds iff $\rho = \sigma$
- $\lim_{\alpha\to 1}\tilde{D}_\alpha(\rho||\sigma) = D(\rho||\sigma)$
- $\tilde{D}_\alpha$ is monotonically increasing in α for $\alpha \in (0,1)\cup(1,\infty)$

The proof can be found in[1,3]

3 Main result

Making use of the property we review in the previous section, we can provide the proof the following theorem.

Theorem 3.1. *Given two density matrix ρ,σ. The following inequality holds for $\alpha \in (1,2]$.*

$$\operatorname{Tr}\rho(\log\frac{\rho+\sigma}{2} - \log\sigma) \leq \frac{1}{\alpha-1}\operatorname{Tr}(\sigma^{\frac{1-\alpha}{2\alpha}}\rho\sigma^{\frac{1-\alpha}{2\alpha}})^\alpha, \forall\alpha\in(1,2]$$

At first glance, we try to connect the left hade-side with quantum relative entropy by restating the problem as $2D(\frac{\rho+\sigma}{2}||\sigma) + D(\sigma||\frac{\rho+\sigma}{2})$, but to no avail. It turns out that it is important to observe

$$\operatorname{Tr}[\rho(\log\frac{\rho+\sigma}{2} - \log\sigma)] \leq D(\rho||\sigma)$$

This is due to the non-negativity of the quantum relative entropy. By doing so, we connected the quantum relative entropy with our problem. Then, it is suffices to prove

$$D(\rho||\sigma) = \lim_{\alpha\to 1}\tilde{D}_\alpha(\rho||\sigma) \leq \frac{1}{\beta-1}\tilde{Q}_\beta(\rho||\sigma), \forall\beta\in(1,2]$$

Proposition 3.2. *Given two density matrix ρ,σ, the following is monotonically increasing in $\alpha \in [1,2]$*

$$\frac{1}{\alpha-1}\tilde{Q}_\alpha(\rho||\sigma) - \frac{1}{\alpha-1}$$

Proof of Theorem 3.1. Let $1 < \alpha < \beta < 2$ and two density matrix ρ,σ. From Proposition 3.2, we have $\frac{1}{\alpha-1}\log\tilde{Q}_\alpha - \frac{1}{\alpha-1} \leq \frac{1}{\beta-1}\log\tilde{Q}_\beta - \frac{1}{\beta-1}$. It implies that

$$\tilde{Q}_\alpha(\rho||\sigma) \leq \frac{\beta-\alpha}{\beta-1} + \frac{\alpha-1}{\beta-1}\tilde{Q}_\beta(\rho||\sigma) \leq 1 + \frac{\alpha-1}{\beta-1}\tilde{Q}_\beta(\rho||\sigma) \leq e^{\frac{\alpha-1}{\beta-1}\tilde{Q}_\beta(\rho||\sigma)}$$

Taking logarithm on both side, we will get

$$\frac{1}{\alpha-1}\log\tilde{Q}_\alpha(\rho||\sigma) \leq \frac{1}{\beta-1}\tilde{Q}_\beta(\rho||\sigma), \forall 1<\alpha<\beta<2$$

To finish our proof, simply take $\alpha \to 1$. We will get

$$D(\rho||\sigma) \leq \frac{1}{\beta-1}\tilde{Q}_\beta(\rho||\sigma), \forall\beta\in(1,2]$$

□

This result can be conjectured in a much stricter bound, which we shall now proved.

$$\operatorname{Tr}\rho(\log\rho+\sigma - \log\sigma) \leq \frac{1}{\alpha-1}\operatorname{Tr}(\sigma^{\frac{1-\alpha}{2\alpha}}\rho\sigma^{\frac{1-\alpha}{2\alpha}})^\alpha, \forall\alpha\in(1,2]$$

Proof. Follows from theorem 3.1, it is suffices to show □

4 Conclusion

We proved an inequality related to sandwiched quasi-relative entropy. Also, from the numerical experiments, we have guessed a tighter bound for the sandwiched quasi-relative entropy.

References

[1] M. M. Wilde, A. Winter, and Dong Hyun Yang. Strong converse for the classical capacity of entanglement-breaking and hadamard channels via a sandwiched rényi relative entropy. *Communications in Mathematical Physics*, 331:593–622, 2013.

[2] R. Bhatia, T. Jain, and Y. Lim. Strong convexity of sandwiched entropies and related optimization problems. *Reviews in Mathematical Physics*, 30:1850014–1850014, 2018.

[3] M. Müller-Lennert, F. Dupuis, Oleg Szehr, S. Fehr, and M. Tomamichel. A new generalization and some properties. *Journal of Mathematical Physics*, 54, 2013b.